Highlights

GPU-Accelerated DMRG on AMD MI300X:
Systematic Benchmarking and Lessons from Failed Optimizations

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- Systematic benchmark: 11 DMRG implementations, 3 physics models, L=8–128, χ=20–256.
- At $\chi \leq 50$, a single CPU core outperforms the 304-CU MI300X in 93% of cases.
- CPU LAPACK SVD consumes 97–98% of per-sweep time at χ=256, capping GPU speedups.
- Newton–Schulz, Block-Davidson, Chebyshev filtering: all fail at $\chi \leq 256$.
- Randomized SVD wins 5–6× only in 2-site DMRG; HIP graph capture wins 1.2–1.8× in parallel DMRG.

GPU-Accelerated DMRG on AMD MI300X: Systematic Benchmarking and Lessons from Failed Optimizations

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Abstract

We present a systematic performance study of eleven density matrix renormalization group (DMRG) implementations, ranging from Python/CPU to hand-tuned GPU kernels in HIP/C++ targeting the AMD Instinct MI300X. The implementations span single-site and two-site DMRG, real-space parallel DMRG with domain decomposition, and an additive two-level parallel variant (A2DMRG), tested across three physics models: the Heisenberg spin-$\frac{1}{2}$ chain ($d = 2$, real), a Josephson junction array ($d = 5$, complex), and the transverse-field Ising model at criticality ($d = 2$, real). Benchmarks cover system sizes $L = 8$–128 and bond dimensions $\chi = 20$–256 on a single MI300X GPU with 192 GB HBM3.

The GPU implementations achieve speedups over optimized CPU code (quimb with OpenBLAS) that depend on both bond dimension and system size: at $\chi = 128$ the GPU wins 100% of short-chain ($L \leq 20$) configurations but only 50% at $L \geq 100$, while at $\chi \leq 50$ the CPU wins all medium-to-large systems ($L \geq 32$). In 27 of 29 CPU-win configurations (93%), a *single* CPU core already outperforms the 304-compute-unit MI300X—the GPU loses not to parallelized CPU BLAS, but to kernel launch latency on small matrices. Two-site DMRG converges 3.3× faster than single-site at large system sizes ($L = 100$, $\chi = 128$: 53 s vs. 177 s).

We report extensive negative results on algorithmic optimizations designed to exploit the MI300X's BLAS-3 throughput. Newton–Schulz iterative polar decomposition, intended to replace SVD with GEMM-heavy iterations, achieves 0% win rate across 50 configurations and diverges numerically at $\chi \geq 128$. Chebyshev-filtered subspace iteration, designed for computing many eigenvalues in density functional theory, is 1.9–11× slower than Lanczos

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for DMRG's single ground state. Cross-segment batched GEMM dispatch is slower in 18 of 19 configurations tested. Profiling reveals the root cause: CPU LAPACK SVD consumes 97–98% of per-sweep wall time at $\chi = 256$, and iterative GPU replacements cannot match LAPACK's optimized QR-iteration algorithm at these matrix sizes. All source code, benchmark scripts, and raw data are publicly available.

Keywords: DMRG, GPU computing, tensor networks, AMD MI300X, performance benchmarking, matrix product states

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PROGRAM SUMMARY

Program Title: `dmrg-implementations`
CPC Library link to program files: (to be added on acceptance)
Developer's repository link:
`https://github.com/yye00/dmrg-implementations`
Code Ocean capsule: (not used)
Licensing provisions: MIT
Programming language: C++17 (∼12 kLOC), Python 3 (∼6 kLOC), MPI (mpi4py) for A2DMRG; GPU code in HIP/C++ targeting ROCm 7.2
External libraries/dependencies: rocBLAS, rocsolver, HIP (ROCm 7.2+); OpenBLAS 0.3.28+ (source-built), LAPACK; quimb 1.8, cotengra, NumPy; mpi4py for parallel Python
Supplementary material: none

Nature of problem: Ground-state computation of one-dimensional quantum lattice models via the density matrix renormalization group (DMRG). Per-bond work is dominated by tensor contractions (GEMM, $O(\chi^3 d^2)$) and singular value decompositions, with moderate bond dimensions $\chi \in [20, 256]$ straddling the CPU–GPU crossover.

Solution method: Eleven DMRG variants spanning single-site and two-site sweeps, domain-decomposition parallel DMRG [1], an additive two-level variant (A2DMRG) [2], and three algorithmic alternatives designed to increase BLAS-3 utilization (Newton–Schulz polar decomposition, Block-Davidson eigensolver, Chebyshev-filtered subspace iteration). All GPU variants share a common env-var-gated ablation framework (six micro-optimizations: DEVICE_K, LANCZOS_GRAPH, RSVD, SPARSE_MPO, FUSE_LANCZOS, D_PAD) so each flag's effect can be measured in isolation on the same binary. Benchmarks run across three

physics models (Heisenberg chain, Josephson-junction array, transverse-field Ising) at system sizes up to $L = 128$.

Additional comments: All results in this paper were generated on the AMD Instinct MI300X (gfx942, ROCm 7.2). Each result JSON embeds git commit SHA, binary SHA-256, runtime CLI, and environment variable snapshot for reproducibility.

1. Introduction

The density matrix renormalization group (DMRG) [3, 4] is the method of choice for computing ground states of one-dimensional quantum many-body systems. In the modern tensor network formulation [5, 6], DMRG variationally optimizes a matrix product state (MPS) by sweeping through the chain and solving local eigenvalue problems at each bond. The computational cost is dominated by tensor contractions (matrix–matrix multiplications) and singular value decompositions (SVDs), both of which scale as $O(\chi^3 d^2)$ per bond, where χ is the bond dimension and d is the local Hilbert space dimension.

Graphics processing units (GPUs) are natural candidates for accelerating DMRG, since the tensor contractions map directly to general matrix–matrix multiplication (GEMM), the workload for which modern GPUs achieve near-peak throughput. The AMD Instinct MI300X, with 304 compute units, 192 GB of HBM3 memory providing 5.3 TB/s bandwidth, and matrix fused multiply-add (MFMA) units delivering 81.7 TFLOPS of FP64 matrix throughput, represents a compelling target for the GEMM-dominated inner loop. Prior work has explored GPU acceleration of DMRG and related tensor network methods [7], typically focusing on NVIDIA hardware with CUDA, but a systematic study of when GPU acceleration helps—and when it does not—has been lacking.

Two questions are open. First, at what problem sizes does GPU acceleration become worthwhile? The practical performance depends on a complex interplay of bond dimension χ (which determines GEMM sizes), local dimension d (which determines batch counts), kernel launch overhead, and the SVD bottleneck, making the answer non-obvious a priori. Second, can algorithmic modifications designed to increase GEMM utilization extend the favorable regime to smaller bond dimensions?

We address both questions through a systematic benchmarking study spanning multiple CPU and GPU DMRG implementations across three physics

models at system sizes $L = 8$–128 and bond dimensions $\chi = 20$–256. We find a narrow window of GPU advantage at large χ and small L, and show through systematic negative results that this window cannot be widened at moderate bond dimensions by several algorithmic modifications designed to exploit BLAS-3 throughput—all of which failed. Remarkably, in 27 of 29 configurations where the CPU wins, a *single* CPU core already outperforms the 304-compute-unit MI300X; the GPU loses not to parallelized CPU BLAS, but to kernel launch latency on matrices small enough to fit in L2 cache. These negative results delineate a boundary between regimes where massive GPU parallelism adds value and regimes where a single CPU core's low-latency sequential execution cannot be beaten.

The paper is organized as follows. Section 2 describes the DMRG algorithm and its GPU implementation, with emphasis on the operations that motivate GPU porting. Section 3 details the algorithmic variants, framing each as a hypothesis about how to shift computation toward the GPU's strengths. Section 4 specifies the hardware, software, and benchmark methodology. Section 5 presents the performance results. Section 6 analyzes the failures and identifies the SVD bottleneck as the root cause. Section 7 summarizes our findings.

2. Methods

This section describes the DMRG algorithm and its GPU implementation. The formalism is standard [5], but we present it with attention to the computational primitives at each step, since the cost structure of these primitives—which operations are BLAS-3, which are BLAS-2, and which are factorizations—determines where GPU acceleration can and cannot help.

2.1. Tensor trains and matrix product states

We consider a one-dimensional quantum lattice of L sites with open boundary conditions. The Hilbert space has dimension d^L, where d is the local basis size (e.g., $d = 2$ for spin-$\frac{1}{2}$), making exact diagonalization intractable beyond $L \approx 20$. The key insight enabling DMRG is the *area law* of entanglement [8]: ground states of gapped local Hamiltonians in one dimension have entanglement entropy bounded by a constant independent of system size, which means they can be efficiently represented by matrix product states (MPS) with moderate bond dimension χ.

An MPS (equivalently, a tensor train in the numerical mathematics literature [9]) parametrizes the d^L-dimensional state vector as a product of L three-index tensors:

$$|\psi\rangle = \sum_{\sigma_1,\ldots,\sigma_L} A_1^{\sigma_1} A_2^{\sigma_2} \cdots A_L^{\sigma_L} \, |\sigma_1 \cdots \sigma_L\rangle, \tag{1}$$

where each $A_i^{\sigma_i}$ is a $\chi_{i-1} \times \chi_i$ matrix (with $\chi_0 = \chi_L = 1$) and $\sigma_i = 1, \ldots, d$ labels the local basis states. The bond dimension $\chi = \max_i \chi_i$ controls the expressiveness of the ansatz: $\chi = 1$ yields a product state, while $\chi = d^{L/2}$ can represent any state exactly. For ground states of local Hamiltonians, $\chi = O(\mathrm{poly}(L))$ typically suffices for fixed accuracy, giving an exponential compression from d^L to $O(L\chi^2 d)$ parameters.

2.2. Canonical forms

Maintaining the MPS in canonical form is essential for numerical stability and for the local eigenvalue problem that drives DMRG. It also introduces the first bottleneck relevant to GPU porting: the SVD-based truncation that maintains canonical form will prove to be the dominant computational cost (Section 5.6).

An MPS is in *left-canonical form* up to site k if each tensor A_i ($i < k$) satisfies $\sum_\sigma A_i^{\sigma\dagger} A_i^\sigma = I$, i.e., each tensor is a (partial) isometry when viewed as a matrix from left to right. Similarly, *right-canonical form* from site k means $\sum_\sigma A_i^\sigma A_i^{\sigma\dagger} = I$ for $i > k$. An MPS in *mixed-canonical form* centered at site k is left-canonical for $i < k$ and right-canonical for $i > k$, so that the tensor at site k (the "active" tensor) contains all the variational freedom.

Canonical form is established and maintained by QR or SVD decompositions. To left-canonicalize site i, reshape A_i into a $(\chi_{i-1}d) \times \chi_i$ matrix and compute the thin QR factorization $A_i = QR$; replace $A_i \leftarrow Q$ and absorb R into A_{i+1}. To truncate the bond dimension (as needed after two-site optimization), SVD is used instead: $A_i = USV^\dagger$, truncate to the $\chi_{\max}$ largest singular values, and split the factors between sites i and $i+1$. The truncation error $\epsilon = \sqrt{\sum_{k>\chi_{\max}} s_k^2}$ is bounded by the discarded singular values, providing a controlled approximation.

2.3. Matrix product operators

The Hamiltonian H is represented as a matrix product operator (MPO) [5, 6]:

$$H = \sum_{\boldsymbol{\sigma},\boldsymbol{\sigma}'} W_1^{\sigma_1\sigma_1'} W_2^{\sigma_2\sigma_2'} \cdots W_L^{\sigma_L\sigma_L'} \, |\boldsymbol{\sigma}\rangle\langle\boldsymbol{\sigma}'|, \tag{2}$$

where each $W_i^{\sigma_i\sigma_i'}$ is a $D_{i-1} \times D_i$ matrix and $D = \max_i D_i$ is the MPO bond dimension. For nearest-neighbor Hamiltonians, D is a small constant: $D = 5$ for the Heisenberg model, $D = 3$ for the transverse-field Ising model, and $D = 4$ for the Josephson junction array. The MPO provides an exact, compact representation of the Hamiltonian that enables efficient computation of expectation values and H_{eff} applications in $O(\chi^2 dD)$ operations per bond.

2.4. The DMRG sweep

The density matrix renormalization group (DMRG) [3, 4] variationally optimizes the MPS by sweeping through the chain. In the modern formulation [5], a left-to-right sweep followed by a right-to-left sweep constitutes one "sweep" of the algorithm. At each bond b, the algorithm:

1. Forms the *effective Hamiltonian* H_{eff} by contracting the left environment L_{env} (encoding all sites $< b$), the MPO tensor(s) W_b, and the right environment R_{env} (encoding all sites $> b$) into a linear operator acting on the active tensor(s) at bond b.
2. Solves the *local eigenvalue problem*

$$H_{\text{eff}}|\theta\rangle = E|\theta\rangle \tag{3}$$

   for the ground-state energy E and optimal local tensor θ, using a Krylov eigensolver (Lanczos).
3. *Restores canonical form*: decomposes the updated tensor via SVD or QR and shifts the canonical center to the next bond.
4. *Updates the environment*: incorporates the newly optimized tensor into L_{env} (left sweep) or R_{env} (right sweep) for the next bond.

The environment tensors L_{env} and R_{env} are rank-3 tensors of dimension $\chi \times D \times \chi$ that encode the contraction of all MPS and MPO tensors to the left (or right) of the active bond. They are updated incrementally as the sweep progresses, at a cost of $O(\chi^2 dD + \chi^2 D^2)$ per bond. The total cost per sweep is $O(L\chi^3 d^2 D)$ for the eigensolve (dominated by the H_{eff} matrix–vector product) plus $O(L\chi^3 d)$ for the SVD/QR step.

Of these operations, steps 1 and 4 are tensor contractions that map to GEMM—precisely the workload where the MI300X's 81.7 TFLOPS matrix throughput should dominate. Step 2 is a sparse Krylov iteration (BLAS-1/2 dominated), and step 3 is a dense factorization. The GPU porting strategy is therefore to maximize the fraction of time spent in steps 1 and 4. Whether this fraction is large enough to overcome launch overhead at moderate χ is the central empirical question of this paper; Section 5 reports the answer.

In *single-site* DMRG, one MPS tensor is optimized at a time, giving an eigensolve on a vector of dimension χd. The bond dimension is fixed throughout the sweep, which is efficient but can trap the optimization in local minima if the initial bond dimension profile is poor.

2.5. Two-site DMRG

In two-site DMRG [10], pairs of adjacent MPS tensors are optimized simultaneously. The two-site tensor θ has dimensions $\chi_L d \times d\chi_R$, and after optimization it is split by SVD:

$$\theta = USV^\dagger, \tag{4}$$

where S is truncated to the $\chi_{\max}$ largest singular values. This allows the bond dimension to *adapt* during the sweep, which is critical for convergence from a random initial state: new Schmidt coefficients can appear that were absent in the initial MPS. The cost per bond is $O(\chi^2 d^2 D)$ for the eigensolve and $O(\chi^3 d^2)$ for the SVD, both scaling with d^2 rather than d compared to single-site.

We precompute fused MPO tensors $\widetilde{W}[b] = W_L[b] \otimes W_R[b]$ for each bond b, so the two-site H_{eff} application uses the same three-step GEMM pattern as single-site but with $d \to d^2$. The larger effective dimension increases the GEMM sizes, which should favor the GPU—but also increases the SVD cost quadratically, which may negate the advantage. Section 5.2 quantifies this tradeoff.

2.6. GPU implementation

The GPU implementation is where the promise of GPU acceleration is realized—or not. The design goal was to maximize the BLAS-3 fraction of the inner loop: every tensor contraction is a GEMM, and the only non-GEMM operations are the Lanczos iteration's vector arithmetic and the SVD factorization.

All GPU implementations target the AMD MI300X via HIP/C++ and use rocBLAS [11] for BLAS operations. The code is templated on the scalar type, supporting both `double` (real) and `hipDoubleComplex` (complex) through a `ScalarTraits` dispatch layer.

The application of H_{eff} to a vector, the innermost operation of the Lanczos eigensolver, is implemented as three batched GEMM steps:

1. **Step 1**: $D \times d^2$ batched GEMMs contracting L_{env} slices with θ slices,
2. **Step 2**: One dense GEMM contracting the intermediate result with the fused MPO $\widetilde{W}$,
3. **Step 3**: $d^2 \times D$ batched GEMMs accumulating the contraction with R_{env} slices into the output.

The pointer arrays for batched GEMMs are computed by lightweight GPU kernels (typically < 64 threads), avoiding host-to-device DMA that would otherwise introduce pipeline stalls [12].

The Lanczos eigensolver runs entirely on the GPU using device-pointer mode for all rocBLAS reductions (`dot`, `nrm2`), eliminating implicit stream synchronizations. Only the final tridiagonal eigensolve (a $k \times k$ problem with $k \approx 15$–50) is performed on the CPU via LAPACK `dstev`. SVD is performed on the CPU via LAPACK `dgesvd` (QR-iteration based; or `zgesvd` for complex), as profiling showed this to be faster than GPU rocsolver for $\chi \leq 256$ (see Section 5.6).

This three-step GEMM structure is the operation we hoped would be the GPU's strength. At $\chi = 128$ with $d = 2$, the individual GEMMs are 128×128—large enough to occupy the MI300X's compute units. At $\chi = 50$, they shrink to 50×50, completing in microseconds while each kernel launch costs 5–10 μs. Section 5.1 maps the resulting crossover surface in (L, χ) space.

2.7. Parallel DMRG

Serial GPU DMRG is limited by the crossover boundary described above: below a certain χ, the per-bond GPU overhead exceeds the CPU's. Parallel DMRG was motivated by the hope that segment parallelism could amortize this per-bond overhead across multiple independent sweeps, extending the GPU-favorable regime to smaller bond dimensions.

Our parallel DMRG implementation follows the real-space decomposition of Stoudenmire and White [1]. The L-site chain is partitioned into P seg-

ments, each assigned to an independent HIP stream and rocBLAS handle. The algorithm proceeds in three phases:

1. **Warmup**: Full-chain *single-site* DMRG sweeps on a single stream to establish an initial converged MPS at fixed bond dimension χ. Single-site is used because the eigensolver problem is smaller (χd vs. χd^2 for two-site), making warmup cheaper.
2. **Outer loop**: Parallel two-site segment sweeps (one `std::thread` per segment, each on its own HIP stream) using staggered directions, followed by boundary merge-and-optimize coupling via the Stoudenmire $V = \Lambda^{-1}$ prescription.
3. **Polish**: Full-chain *single-site* sweeps until convergence. Single-site is again used for the polish phase, as the bond dimension is already established and single-site sweeps are cheaper per bond.

The coupling step is essential: segment sweeps disrupt the global entanglement structure of the MPS, and only boundary coupling or full-chain resweeping restores the correct ground state. Experiments with boundary-only coupling sweeps (optimizing only sites near segment boundaries) failed to converge (energy errors > 8), confirming that coupling at every boundary is necessary. Section 5.3 reports that this serial coupling requirement limits parallel DMRG to being slower than serial two-site DMRG at all tested sizes on a single GPU.

3. Algorithmic variants

The baseline GPU DMRG of Section 2 wins at high χ and small L but loses at moderate bond dimensions, where kernel launch overhead dominates the small GEMMs and CPU SVD consumes 97–98% of wall time (Section 5.6). This section describes five algorithmic modifications, each motivated by a specific hypothesis about how to shift computation toward the GPU's strengths. All five hypotheses were tested and all five were rejected by the benchmark data; the details of each failure are reported in Section 5.4.

3.1. Newton–Schulz polar decomposition

Hypothesis: Replacing the BLAS-2-dominated SVD with GEMM iterations should shift the dominant computation to the GPU's strength, reducing or eliminating the CPU SVD bottleneck.

The Newton–Schulz iteration computes the unitary polar factor U of a matrix $A = UP$ (where P is positive semidefinite) via the recurrence [13, 14]

$$X_0 = A/\|A\|_F, \qquad X_{k+1} = \tfrac{1}{2} X_k (3I - X_k^\dagger X_k). \tag{5}$$

This converges cubically to the unitary polar factor, with a maximum of 30 iterations (each consisting of two GEMMs) and convergence criterion $\|X^\dagger X - I\|_F < 10^{-10}$. After convergence, the singular values are obtained from the eigendecomposition of $P = U^\dagger A$, performed on the CPU.

The intent was to replace the LAPACK SVD call with ∼10–20 GPU GEMMs, shifting the computation from a BLAS-2-dominated factorization to pure BLAS-3. For single-site DMRG moving left-to-right, the MPS tensor θ is tall ($\chi_L d \times \chi_R$) and Newton–Schulz applies directly. For the reverse direction, θ is wide and we fall back to LAPACK SVD. Section 5.4 reports that this variant achieves a 0% win rate across 50 configurations and diverges numerically at $\chi \geq 128$.

3.2. Block-Davidson eigensolver

Hypothesis: BLAS-3 subspace projections should outperform the BLAS-1/2-dominated Lanczos iterations for the local eigenvalue problem.

The Block-Davidson method [15, 16] replaces Lanczos for the local eigenvalue problem (3). It maintains a subspace of block size $b = 4$ trial vectors, projects H_{eff} into the subspace using GEMM, solves the small projected eigenproblem via LAPACK `dsyev`, and expands the subspace with preconditioned residual corrections. The maximum subspace dimension is 32, after which the method restarts from the best eigenvector.

The motivation was that Lanczos is BLAS-1/2 dominated (one matrix–vector product plus dot products and axpy operations per iteration), while Davidson's subspace projection is a BLAS-3 operation (GEMM of the subspace basis with the H_{eff} application results). However, this trades a lower iteration count for higher per-iteration overhead from subspace management; Section 5.4 shows the tradeoff is unfavorable.

3.3. Chebyshev-filtered subspace iteration

Hypothesis: Polynomial filtering can reduce eigenvalue iteration count by suppressing unwanted spectral components, and the filter application is a sequence of matrix–vector products that map well to the GPU.

Chebyshev-filtered subspace iteration (CheFSI) [17] uses a polynomial filter to suppress unwanted eigenvalue components. Given spectral bounds

$[\lambda_{\min}, \lambda_{\max}]$ from a short Lanczos run (10 steps), a degree-p Chebyshev polynomial T_p is applied via the three-term recurrence

$$y_{k+1} = 2\varphi(H)y_k - y_{k-1}, \qquad \varphi(H) = \frac{2H - (\lambda_{\max} + \lambda_{\min})I}{\lambda_{\max} - \lambda_{\min}}, \tag{6}$$

where each application of $\varphi(H)$ requires one H_{eff} matrix–vector product. The energy is extracted via the Rayleigh quotient after n_{outer} filter applications.

CheFSI was designed for density functional theory (DFT) calculations where many eigenvalues are needed simultaneously [17]. We tested it for DMRG's single ground state search with polynomial degree $p = 15$ and $n_{\text{outer}} = 20$. The fundamental mismatch—a method designed for many eigenvalues applied to a single-eigenvalue problem—foreshadows the 1.9–11× slowdown reported in Section 5.4.

3.4. Cross-segment batched GEMM

Hypothesis: Batching GEMM calls across parallel segments reduces per-kernel launch overhead, the dominant cost at moderate χ.

For parallel DMRG, we implemented a lock-step sweep mode that batches GEMM calls across segments. Instead of each segment independently dispatching GEMMs on its own stream, segments with matching bond dimensions are grouped and their GEMMs are dispatched as a single batched-GEMM call (`rocblas_gemm_batched`). This was intended to reduce per-kernel launch overhead by combining P small GEMMs into one batched dispatch. Section 5.4 reports that the lock-step synchronization serializes other operations, making this slower in 18 of 19 configurations.

3.5. Additive two-level parallel DMRG (A2DMRG)

Hypothesis: Additive parallelism avoids the serial coupling bottleneck of real-space decomposition by updating all sites independently and combining results via a coarse-space correction.

We implemented the additive two-level parallel DMRG (A2DMRG) algorithm of Grigori and Hassan [2] in Python with MPI parallelization. Unlike the real-space decomposition of Section 2.7, which partitions the chain into segments, A2DMRG updates *all* sites independently and combines the results via a coarse-space correction.

The algorithm proceeds in five phases per sweep:

1. **Orthogonal decompositions**: Construct d i-orthogonal MPS forms, one centered at each site.

2. **Parallel local micro-steps**: Solve the local eigenvalue problem (3) independently at every site, producing L candidate tensors. These are embarrassingly parallel across P processors.
3. **Coarse-space correction**: Form the $(d+1)\times(d+1)$ projected Hamiltonian and overlap matrices H_{coarse}, S_{coarse} in the subspace spanned by the d candidates plus the current MPS, and solve the generalized eigenvalue problem to find the optimal linear combination.
4. **Linear combination**: Assemble the updated MPS as the coarse-space eigenvector applied to the candidate tensors. This produces a bond dimension of up to $(L-1)\times\chi$.
5. **Compression**: Project back to the target bond dimension χ via TT-SVD (sequential truncated SVD from left to right).

The algorithm is an additive Schwarz method with coarse-space correction, analogous to the relationship between Jacobi (additive) and Gauss–Seidel (multiplicative) iterations for linear systems. Standard DMRG's sequential sweep is the multiplicative variant and converges faster per sweep, but the additive structure allows all local problems to be solved in parallel.

Our implementation includes three optimizations not in the original paper: (i) incremental environment caching (reducing the per-sweep cost from $O(L^2\chi^2 D)$ to $O(L\chi^2 D)$), (ii) a pure numpy hot path replacing quimb tensor objects in the sweep phases, and (iii) configurable warm-up sweeps (serial DMRG2 via quimb) before entering the parallel A2DMRG loop.

To our knowledge, this is the first independent implementation of A2DMRG. The original paper [2] tested only quantum chemistry Hamiltonians (hydrogen chains H_6–H_{12}, C_2, N_2) with system sizes $d \leq 24$ using an in-house Julia implementation that is not publicly available. Our implementation extends the algorithm to lattice spin models (Heisenberg, Josephson, TFIM) and tests larger system sizes (L up to 64). Section 5.4 reports that accuracy degrades beyond $L \approx 20$ and that serial warm-up renders the parallel phase redundant—a paradox that Section 6 analyzes in detail.

3.6. GPU micro-optimization framework

The five algorithmic variants above are large-grain hypotheses: each replaces an entire kernel (SVD, eigensolver, sweep parallelism) with a different algorithm. A complementary question is whether smaller, BLAS-3-preserving optimizations of the existing baseline—targeting kernel launch overhead,

memory traffic, or specific sub-bottlenecks—can move the needle without changing the algorithm.

To answer this systematically, we built an env-var-gated ablation framework. Each of six independent optimizations is a runtime flag (`DMRG_GPU_OPT_<NAME>=1`) wired into every `-gpu` and `-gpu-opt` variant via a shared `GpuOpts` struct, so the same binary reproduces baseline numerics with all flags off and exercises any combination otherwise. The six flags target distinct mechanisms:

- **DEVICE_K**: skip the device→host readback of the per-bond truncation rank by always retaining $\chi_{\max}$ singular values; eliminates one stream drain per optimized site.
- **LANCZOS_GRAPH**: capture each variant's `apply_heff` call as a HIP graph keyed on $(\text{site}, \chi_L, \chi_R)$ and replay on subsequent Lanczos iterations; amortizes per-kernel launch overhead across the inner loop.
- **RSVD**: Halko–Martinsson–Tropp randomized SVD for the bond-splitting step, replacing `rocsolver_gesvd` with the chain $Y = A\Omega$, $Q = \mathrm{qr}(Y)$, $B = Q^H A$, small SVD of B, $U = Q\,U_{\text{small}}$.
- **SPARSE_MPO**: detect zero rows and columns of the fused MPO matrix at construction time and compact the Step 1 / Step 3 batched GEMMs in `apply_heff` accordingly.
- **FUSE_LANCZOS**: replace the per-iteration rocBLAS axpy/memcpy/scal chain with two fused kernels (one for $w := w - \alpha v_i - \beta v_{i-1}$, the other for $v_{i+1} := w/\beta_i$).
- **D_PAD**: round the MPO bond dimension up to the next multiple of 8 (zero-filling padded rows/columns of W) for MFMA-friendly GEMM shapes.

A seventh flag, `LANCZOS_FIXED` (a fixed-iteration Lanczos eliminating mid-loop convergence-check syncs), was included in early ablation runs but consistently regressed performance by 1.4–2.5× across all tested variants and was removed from the framework. The fixed-count Lanczos doubles the number of `apply_heff` calls relative to the adaptive baseline, and the per-call cost dominates the saved synchronization.

The ablation script (`benchmarks/bench_dmrg_gpu_ablate.py`) runs each variant through baseline (all flags off), one-at-a-time (each flag in isolation), leave-one-out (all flags except one), and all-on configurations, with $N = 3$ repeats per cell and a built-in correctness gate at $|\Delta E| < 5 \times 10^{-10}$. Results are reported per-flag in Section 5.5.

4. Computational details

4.1. Hardware and software

All GPU benchmarks were run on a single AMD Instinct MI300X accelerator (gfx942 architecture) with 304 compute units and 192 GB HBM3 memory providing 5.3 TB/s bandwidth. The peak FP64 throughput is 81.7 TFLOPS for matrix operations (MFMA instructions) and 40.9 TFLOPS for vector operations. The host CPU is an AMD EPYC processor.

We deliberately restrict the GPU comparison to the MI300X. The natural NVIDIA counterpart, the H100, supports FP64 with significantly different hardware semantics: 33.5 TFLOPS FP64 vector and 67 TFLOPS FP64 tensor (the latter using a fused-MMA path with internal accumulation rules that differ from a fully native FP64 matrix unit) versus the MI300X's 81.7 TFLOPS native FP64 MFMA. Our preliminary H100 runs (in `benchmarks/paper_results/h100/`, 18 matching configurations across two variants) show H100 wall times that are 0.18–1.65× those of the MI300X depending on χ and L, but disentangling native-vs-emulated FP64 contributions from algorithmic effects requires per-kernel profiling that is out of scope here. A controlled cross-vendor comparison is left to future work; the present study reports only single-vendor results to keep all measured speedups attributable to the algorithmic and software changes under investigation.

The software environment consists of ROCm 7.2, rocBLAS (for GEMM), rocsolver (for GPU SVD, used in comparisons), and LAPACK via OpenBLAS 0.3.28 compiled from source. The system-provided OpenBLAS 0.3.20 was found to contain a bug in `dgesvd` that produced incorrect singular vectors; we compiled 0.3.28 as a workaround. All GPU code was compiled with `hipcc` (clang-based) targeting `gfx942`.

CPU benchmarks used the quimb tensor network library [18] (version 1.8) with cotengra [19] for automatic contraction path optimization, backed by OpenBLAS with thread counts of 1, 2, 4, 8, and 12.

4.2. Benchmark methodology

Each benchmark configuration specifies the implementation, physics model, system size L, bond dimension χ, and number of sweeps (or outer iterations for parallel DMRG). All runs use open boundary conditions. Wall times are measured from program start to convergence or sweep limit, including all initialization. Reported times are from the internal "Total wall time" timer, which excludes only process startup.

Correctness is validated against exact diagonalization energies for small systems ($L = 4$–8) and against converged quimb reference energies for larger systems, with a pass threshold of $|\Delta E| < 10^{-10}$. A total of 604 benchmark configurations were run for the main comparison (all implementations $\times$ all models $\times$ all sizes), plus 108 configurations for the GPU optimization scaling study.

For CPU benchmarks, we report the best wall time across all thread counts (1, 2, 4, 8, 12) for each configuration, as the optimal thread count varies with problem size.

4.3. Physics models

We benchmark three Hamiltonians that exercise different aspects of the DMRG implementation:

Heisenberg spin-$\frac{1}{2}$ chain..

$$H = \sum_{i=1}^{L-1} \left(S_i^x S_{i+1}^x + S_i^y S_{i+1}^y + S_i^z S_{i+1}^z \right), \tag{7}$$

with $d = 2$, real arithmetic, and MPO bond dimension $D = 5$. This is the standard DMRG benchmark [3], with the exact ground state energy density approaching $e_0 = 1/4 - \ln 2 \approx -0.4432$ per bond in the thermodynamic limit [20].

Josephson junction array..

$$H = -\frac{E_J}{2} \sum_i \left(e^{i\varphi_{\text{ext}}} e^{i\hat{\varphi}_i} e^{-i\hat{\varphi}_{i+1}} + \text{h.c.} \right) + E_C \sum_i \hat{n}_i^2, \tag{8}$$

with $E_J = 1.0$, $E_C = 0.5$, external flux $\varphi_{\text{ext}} = \pi/4$, and charge basis truncation $n_{\max} = 2$ giving $d = 5$ and complex arithmetic [21]. The MPO bond dimension is $D = 4$. The nonzero external flux breaks time-reversal symmetry, requiring full complex-valued MPS and MPO tensors.

Transverse-field Ising model (TFIM)..

$$H = -J \sum_i Z_i Z_{i+1} - h \sum_i X_i, \tag{9}$$

with $h/J = 1.0$ at the quantum critical point [22], $d = 2$, real arithmetic, and $D = 3$. At criticality, the entanglement entropy grows logarithmically with subsystem size, $S \sim (c/6) \ln L$ with central charge $c = 1/2$, providing a stress test for bond dimension convergence.

5. Results

We present the benchmark results in a sequence that traces a natural line of inquiry: Where does the GPU first beat the CPU? Which GPU strategy is best? Can parallelism or algorithmic cleverness widen the GPU's advantage? And what single bottleneck explains why so many attempts fail?

5.1. Baseline crossover: when does the GPU win?

Table 1 summarizes the win rates between GPU and CPU implementations across all converged benchmark configurations, stratified by bond dimension. A "win" is assigned to the fastest implementation for each (L, χ, model) configuration, considering all GPU implementations and the best CPU result across thread counts.

Table 1: Win rates by bond dimension. D1-GPU: single-site GPU DMRG; D2-GPU: two-site GPU DMRG; Q-D1, Q-D2: quimb single-/two-site with best thread count. All Newton–Schulz/Davidson variants achieve 0% win rate and are omitted.

χ	D1-GPU	D2-GPU	Q-D1	Q-D2
20	25.0%	25.0%	37.5%	12.5%
50	26.7%	0%	60.0%	13.3%
128	80.0%	0%	20.0%	0%

At $\chi = 20$–50, CPU implementations win the majority of configurations. A striking detail clarifies why: of all CPU wins across all χ values, 93% are achieved with single-threaded OpenBLAS ($t = 1$). The individual GEMMs at $\chi = 50$ (matrices of size $\sim 50 \times 50$) fit entirely in CPU L2 cache, and call overhead dominates over arithmetic; additional threads add synchronization cost without meaningful parallelism. The GPU faces the same small-matrix

problem from the opposite direction: kernel launch overhead ($\sim$5–10 μs per launch) across hundreds of launches per Lanczos iteration outweighs the compute. Both the CPU and GPU are bottlenecked by overhead, not by FLOPS—but the CPU's overhead is smaller.

The crossover depends on *both* bond dimension and system size. Table 2 stratifies the GPU vs. CPU win rate by L range and χ. At small L (≤ 20), the GPU already wins 67% of configurations even at $\chi = 20$, because short chains require few sweeps and the per-bond GPU advantage dominates. At medium L (32–64), the CPU wins every configuration at $\chi \leq 50$—the increased number of bonds amplifies kernel launch overhead—and the GPU recovers only at $\chi = 128$ (71%). At large L (≥ 100), even $\chi = 128$ yields only a 50% GPU win rate, because the many-sweep convergence behavior favors the CPU's lower per-iteration overhead at moderate matrix sizes.

Table 2: GPU vs. CPU win rates stratified by system size range and bond dimension. Each cell shows GPU wins / total configurations (win rate). CPU wins are the complement.

L range	$\chi = 20$	$\chi = 50$	$\chi = 128$
8–20	4/6 (67%)	4/6 (67%)	6/6 (100%)
32–64	0/2 (0%)	0/7 (0%)	5/7 (71%)
≥ 100	—	0/2 (0%)	1/2 (50%)

Among CPU implementations, quimb single-site DMRG (quimb-dmrg1) dominates at all system sizes and bond dimensions, winning 83–100% of configurations across all (L, χ) cells (Table 3). The only exception is at large L (≥ 64) with $\chi = 50$, where quimb two-site DMRG wins half of the configurations—the same regime where two-site convergence advantages appear. Our custom Python PDMRG never wins.

Table 3: CPU implementation win rates by system size and bond dimension. Only converged runs included. Cells show quimb-dmrg1 / quimb-dmrg2 wins (total configs).

L range	$\chi = 20$	$\chi = 50$	$\chi = 128$
8–20	5/1 (6)	6/0 (6)	5/1 (6)
32–64	2/0 (2)	6/1 (7)	5/0 (5)
≥ 100	—	1/1 (2)	2/0 (2)

The crossover picture is clear: the GPU needs $\chi \geq 128$ to compete, and

even then system size erodes the advantage. The next question is which GPU strategy makes the most of that narrow window.

5.2. Best GPU strategy: single-site vs. two-site

Within the GPU regime, the choice between single-site (dmrg-gpu) and two-site (dmrg2-gpu) depends on both system size and bond dimension (Table 4). At $\chi \geq 50$, single-site wins the majority of configurations at all system sizes (71–100%). Single-site's advantage comes from the smaller local eigensolve (χd vs. χd^2), which is particularly pronounced for the Josephson model ($d = 5$, so $d^2 = 25$).

At $\chi = 20$, the picture reverses for medium-to-large systems: two-site wins 100% of $L = 32$–64 configurations because adaptive bond dimension growth converges in 2–3 sweeps vs. 5–10 for single-site, and the per-bond cost penalty from d^2 is modest at small χ. At large L (≥ 100) with $\chi = 128$, the split is 50/50—two-site's convergence advantage is balanced against its higher per-bond cost when many sweeps are needed.

Table 4: GPU implementation win rates (single-site / two-site) by system size and bond dimension. Only converged runs with both implementations available.

L range	$\chi = 20$	$\chi = 50$	$\chi = 128$
8–20	3/3 (6)	5/1 (6)	6/0 (6)
32–64	0/2 (2)	5/2 (7)	6/1 (7)
≥ 100	—	1/1 (2)	1/1 (2)

Table 5 quantifies the convergence advantage directly. At $L = 100$, $\chi = 128$, dmrg2-gpu completes in 53 s vs. 177 s for dmrg-gpu (3.3× faster), because two-site optimization converges in 2–3 sweeps by dynamically growing the bond dimension, while single-site requires 5–10 sweeps at fixed χ. The "-opt" variants (Newton–Schulz + Block-Davidson) are uniformly slower, as analyzed in Section 5.4.

The practical recommendation is: use single-site dmrg-gpu at $\chi \geq 50$ for most problems; switch to two-site dmrg2-gpu when convergence is slow ($L \geq 64$ with small d) or when $\chi \leq 20$ and the system is not tiny. Parallel DMRG (pdmrg-gpu) never wins on a single GPU.

Given that the GPU's advantage is confined to a narrow parameter window, can parallelism widen that window by distributing work across segments or processors?

Table 5: Serial GPU wall times (seconds) for Heisenberg OBC. Times include all sweeps to convergence.

L	χ	dmrg-gpu	dmrg2-gpu	dmrg-gpu-opt	dmrg2-gpu-opt
8	32	0.31	0.34	0.68	0.78
32	128	4.21	5.63	5.09	8.87
64	128	24.04	22.83	26.63	30.64
100	128	176.80	53.13	157.13	69.32

5.3. Parallelism: Stoudenmire and A2DMRG

5.3.1. Real-space parallel DMRG (Stoudenmire)

The parallel DMRG (pdmrg-gpu) partitions the chain into P segments with independent sweeps. Table 6 shows wall times for the optimized parallel implementation.

Table 6: Parallel GPU DMRG wall times (seconds), Heisenberg OBC.

L	χ	$P = 2$	$P = 4$
8	32	0.31	—
20	50	2.54	—
32	128	16.46	—
64	128	49.38	78.11
100	128	216.46	196.52

Parallel DMRG is slower than serial two-site DMRG (dmrg2-gpu) at all tested sizes. At $L = 64$, $\chi = 128$, pdmrg-gpu takes 49 s with $P = 2$ segments vs. 23 s for dmrg2-gpu. The overhead comes from three sources: (i) warmup sweeps are serial single-site sweeps, (ii) segment sweeps disrupt the global MPS entanglement structure, requiring expensive full-chain coupling sweeps to restore, (iii) the coupling sweep is itself a full serial sweep.

Amdahl's law limits the potential speedup. At $L = 64$, $\chi = 128$, the serial fraction (warmup + coupling + polish) accounts for 22.1 s out of 24.1 s total, capping the theoretical speedup at $1.22\times$ over serial DMRG even with zero-cost parallel segments.

5.3.2. Additive two-level parallel DMRG (A2DMRG)

The A2DMRG algorithm (Section 3.5) takes a fundamentally different approach: instead of partitioning the chain spatially, it updates *all* sites independently and combines the results via a coarse-space correction. We tested it on the Heisenberg model with $P = 2$ MPI ranks and 2 serial DMRG2 warm-up sweeps, followed by up to 20 A2DMRG iterations. Table 7 shows accuracy and wall times compared to serial quimb DMRG2.

Table 7: A2DMRG vs. serial DMRG2 (quimb) on the Heisenberg chain, $P = 2$, 2 warm-up sweeps. $|\Delta E|$ is the absolute energy error vs. converged reference.

L	χ	$\lvert\Delta E\rvert$	A2DMRG (s)	DMRG2 (s)	Status
8	20	3×10^{-15}	0.1	0.1	PASS
12	20	7×10^{-15}	0.2	0.1	PASS
12	50	7×10^{-15}	0.3	0.2	PASS
16	50	2×10^{-14}	0.7	0.4	PASS
20	50	5×10^{-11}	2.1	0.8	PASS
32	50	2.7×10^{-7}	7.4	5.2	FAIL
32	100	8.9×10^{-9}	35.6	11.4	WARN
48	50	3.1×10^{-6}	15.9	11.4	FAIL
48	100	1.9×10^{-5}	104.3	26.8	FAIL
64	50	5.7×10^{-5}	32.7	17.4	FAIL
64	100	1.9×10^{-5}	198.3	44.4	FAIL

A2DMRG achieves machine-precision accuracy ($|\Delta E| < 10^{-10}$) for $L \leq 20$, matching the system sizes tested in the original paper [2] ($d \leq 24$). However, accuracy degrades rapidly beyond $L \approx 30$: at $L = 64$, errors reach 10^{-5}, and the algorithm is 2–4× slower than serial DMRG2 even with only $P = 2$ ranks.

The degradation has a clear structural cause: the linear combination phase produces bond dimensions up to $(L-1) \times \chi$, which must be compressed back to χ via TT-SVD. At $L = 32$, $\chi = 20$, this is a 31:1 compression ratio, causing significant information loss. The original paper avoids this regime by testing quantum chemistry Hamiltonians with $d \leq 24$ and using a rank-growth strategy that starts with small bond dimensions and gradually increases them, yielding gentler compression ratios in early sweeps.

A diagnostic experiment reveals a deeper issue with the algorithm's practical utility. Table 8 shows how A2DMRG accuracy at $L = 32$ depends on

the number of serial DMRG2 warm-up sweeps.

Table 8: A2DMRG accuracy vs. number of serial DMRG2 warm-up sweeps (Heisenberg $L = 32$, $\chi = 20$, $P = 4$).

Warm-up	$\lvert\Delta E\rvert$	Total time (s)	Note
0	1.27	74.8	Not converged
1	5.7×10^{-3}	0.2	—
2	4.4×10^{-6}	0.3	—
3	1.5×10^{-8}	0.3	—
5	5.2×10^{-12}	0.4	Machine precision
10	2.9×10^{-13}	0.4	—

Each additional warm-up sweep buys 2–3 orders of magnitude in accuracy. By 5 warm-up sweeps, A2DMRG reaches machine precision—but the A2DMRG phase itself then converges in 1–2 iterations taking < 0.1 s, because the serial warm-up has already produced a nearly-converged state. This creates a paradox: sufficient warm-up makes the algorithm accurate but also makes the parallel phase redundant, since serial DMRG2 alone would have converged in the same time.

We also tested A2DMRG on a complex-valued Bose–Hubbard/Josephson Hamiltonian ($d = 5$, $L = 16$), where it failed catastrophically ($|\Delta E| \sim 10^3$). The original paper tests only real-valued quantum chemistry Hamiltonians, and the complex eigensolver and TT-SVD compression path may require additional stabilization for non-Hermitian effective Hamiltonians.

Three important caveats apply to our A2DMRG results. First, the original paper reports theoretical per-processor FLOP counts, not wall-clock times, assuming $d - 1$ processors (one per site); our $P = 2$ tests are in the algorithm's worst operating regime. Second, quantum chemistry Hamiltonians have all-to-all MPO structure that may favor the coarse-space correction more than the constant-D MPOs of lattice models. Third, the rank-growth initialization strategy used in the original paper avoids the large compression ratios that degrade our cold-start results.

Neither form of parallelism widens the GPU's competitive window. Perhaps a different tactic—replacing the algorithms themselves with GPU-friendlier variants—can succeed where parallelism failed.

5.4. Algorithmic variants: can smarter algorithms help?

5.4.1. Newton–Schulz + Block-Davidson: 0% win rate

The Newton–Schulz variants (dmrg-gpu-opt, dmrg2-gpu-opt) lose to their baselines in every converged comparison: 0 wins out of 50 configurations (24 single-site, 26 two-site). Table 9 shows representative results.

Table 9: Algorithmic variant (opt) vs. baseline wall times (seconds). All models, converged runs only. Speedup < 1 means the variant is slower.

Model	L	χ	Type	Baseline	Opt
Heisenberg	16	20	1-site	1.7	3.5
Heisenberg	32	50	2-site	3.5	8.0
Heisenberg	64	20	1-site	16.6	55.1
Josephson	16	50	2-site	2.9	8.4
Josephson	32	256	1-site	18.4	104.6
TFIM	32	128	1-site	9.8	24.2
TFIM	128	50	2-site	28.1	39.3

The slowdown ranges from 1.4× (best case, TFIM $L = 128$ $\chi = 50$) to 5.7× (Josephson $L = 32$ $\chi = 256$).

At $\chi \geq 128$, Newton–Schulz diverges catastrophically for the Heisenberg model. The Frobenius-norm-scaled iteration (5) fails to converge when the condition number of the two-site tensor θ is large, producing MPS tensors with corrupted entries that cause the energy to diverge to $O(10^6)$–$O(10^{20})$. A fallback to LAPACK SVD is triggered when $\|X^\dagger X - I\|_F > 10^{-8}$, but by the time the fallback activates, the accumulated error from previous corrupted truncations has already destabilized the MPS.

5.4.2. Chebyshev-filtered subspace iteration: 1.9–11× *slower*

CheFSI with degree $p = 15$ and $n_{\text{outer}} = 20$ converges to the correct energy ($|\Delta E| < 10^{-13}$) but is dramatically slower than Lanczos:

The root cause is a fundamental mismatch between algorithm and problem. CheFSI applies a degree-p polynomial filter for n_{outer} iterations, requiring up to $p \times n_{\text{outer}} = 300$ matrix–vector products. Lanczos converges in ∼15 iterations for DMRG's single ground state. CheFSI was designed for DFT calculations [17] where many eigenvalues are computed simultaneously,

Table 10: Chebyshev-filtered subspace iteration vs. Lanczos eigensolver (pdmrg-gpu-opt, Heisenberg, $P = 2$ segments).

L	χ	Lanczos (s)	CheFSI (s)	Slowdown
8	32	0.73	1.40	1.9×
20	50	5.27	59.15	11.2×

amortizing the polynomial cost across $O(N_e)$ eigenstates. For a single eigenvalue, Krylov methods are fundamentally more efficient: Lanczos achieves $O(\sqrt{\kappa})$ convergence that no polynomial filter can match.

5.4.3. Cross-segment batched GEMM: slower at 18/19 configurations

The batched sweep mode (Section 3.4) was tested across 19 configurations with $L = 20$–100, $\chi = 50$–256, and $P = 2$–16 segments. Table 11 shows representative results.

Table 11: Cross-segment batched GEMM sweep vs. thread-per-segment baseline (pdmrg-gpu-opt, Heisenberg).

L	χ	P	Baseline (s)	Batched (s)	Speedup
20	50	2	2.35	3.04	0.77×
32	128	2	16.55	231.81	0.07×
32	128	4	15.37	25.56	0.60×
32	256	2	59.91	50.66	**1.18**×
32	256	4	25.86	40.25	0.64×
64	128	2	72.23	timeout	—
100	128	4	—	196.52	—

Only one configuration ($\chi = 256$, $P = 2$) shows a speedup (1.18×). The batched mode fails because: (i) the lock-step sweep serializes BLAS-1 operations (dot, nrm2, axpy) in the Lanczos eigensolver that run concurrently in the thread-per-segment baseline; (ii) segments often have mismatched bond dimensions, preventing effective GEMM batching; (iii) at $\chi = 256$ with only $P = 2$ segments, the individual GEMMs are large enough that batching the pointer array dispatch provides modest savings.

Parallelism failed because serial phases dominate; algorithmic variants failed because they increase total work. These failures—spanning four in-

Table 12: Per-flag median speedup over the variant baseline on Josephson $L = 32$, $n_{\max} = 2$, median of three repeats per cell. Each cell shows $\chi = 128$ / $\chi = 256$. Bold = $> 1.05\times$; em-dash = within $\pm 5\%$ of baseline; n/a = flag inapplicable; † = flag force-disabled at construction (Block-Davidson incompatibility, see text).

Flag	`dmrg-gpu`	`dmrg2-gpu`	`pdmrg-gpu`	`dmrg-gpu-opt`	`dmrg2-gpu-opt`	`pdmrg-gpu-opt`
`DEVICE_K`	–/–	–/1.02	1.01/–	n/a	n/a	–/–
`LANCZOS_GRAPH`	–/–	0.99/0.81	**1.18/1.77**	†	†	1.02/–
`RSVD`	–/–	**5.49/6.29**	–/–	n/a	n/a	n/a
`SPARSE_MPO`	–/–	**1.15/1.31**	–/–	–/–	0.95/–	–/–
`FUSE_LANCZOS`	–/–	–/0.81	–/–	–/–	–/–	–/–
`D_PAD`	–/–	**1.14**/–	–/–	–/–	1.01/–	–/–

dependent strategies—share a common explanation, which profiling reveals directly.

5.5. Per-flag ablation across variants

Before turning to the SVD profile, we present the results of the micro-optimization ablation framework introduced in Section 3.6. The framework lets us measure each flag's effect in isolation across all six `-gpu` and `-gpu-opt` variants on a fixed Josephson-junction array workload (complex128, $L = 32$, $n_{\max} = 2$, three repeats per cell, median wall-time reported).

Two configurations were swept: $\chi = 128$ (apply_heff/SVD-mixed regime) and $\chi = 256$ (SVD-dominated). Table 12 reports the speedup of each flag on its own (*only-X*) relative to the variant's flags-off baseline. A speedup $> 1.05\times$ is highlighted; values within $\pm 5\%$ of unity are reported as "–" (within run-to-run noise). Crashes or correctness violations are marked †.

Figure 1 visualizes the same data as a heatmap. The headline observation is that *four of six flags are flat*—DEVICE_K, SPARSE_MPO (in 1-site), FUSE_LANCZOS, and D_PAD all sit within $\pm 5\%$ of baseline almost everywhere. DEVICE_K eliminates one stream drain per site, but the drained streams are followed by overlapping work whose serialization point migrated rather than disappeared. FUSE_LANCZOS replaces six rocBLAS calls per Lanczos iteration with two custom kernels, but the per-call BLAS-1/2 operations are bandwidth-dominated and the launch savings are dwarfed by the unchanged `apply_heff` time inside each iteration. D_PAD adds zero-padded dimensions to the MPO bond axis to align GEMM shapes with MFMA tile

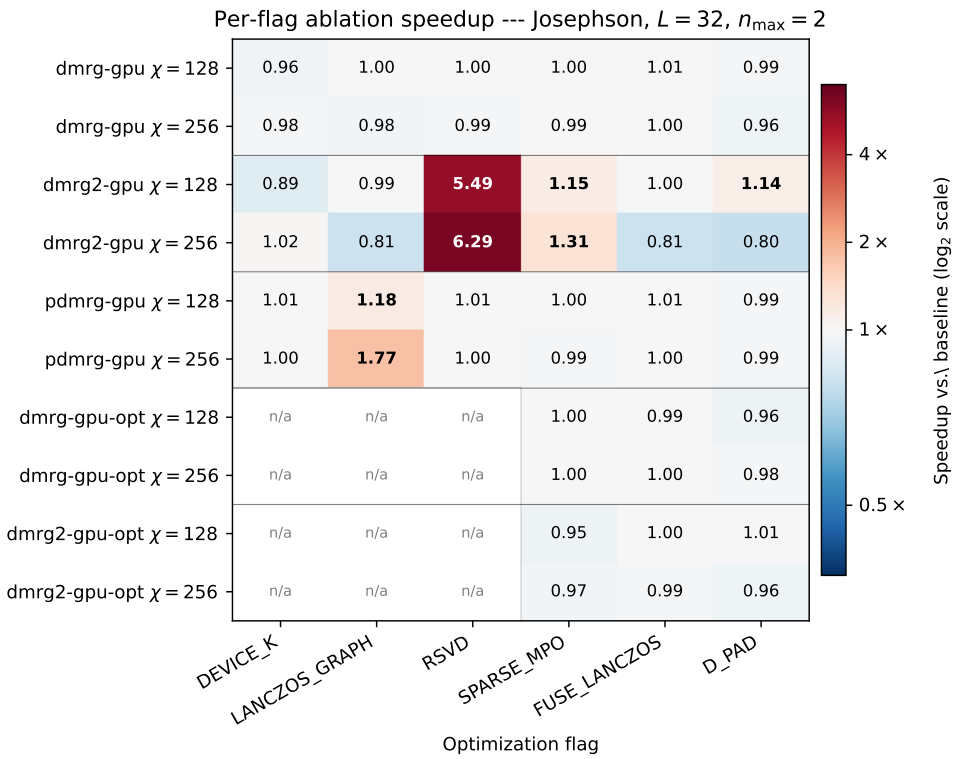

Figure 1: Per-flag speedup over baseline across the six `-gpu` and `-gpu-opt` variants on the Josephson workload. Cells show only-X (single-flag) speedups at $\chi = 128$ and $\chi = 256$; "n/a" marks flag-variant pairs where the flag is inapplicable (no device SVD truncation kernel for DEVICE_K, CPU LAPACK SVD path for RSVD) or constructor-disabled (LANCZOS_GRAPH on Block-Davidson `-gpu-opt`). Two cells exceed 1.5×: RSVD on dmrg2-gpu and LANCZOS_GRAPH on pdmrg-gpu.

sizes; for the tested $D \in \{4, 5\}$ this gives the right alignment in principle but the smallest GEMMs are already memory-bound, not throughput-bound. SPARSE_MPO compacts batched GEMMs to the nonzero columns of the MPO; the Heisenberg and Josephson MPOs are sufficiently dense at the tested sizes that the saved batches do not exceed the bookkeeping cost—except in the 2-site fused MPO where the redundant identity blocks expose enough sparsity to net 15–31%.

The two non-flat findings are more interesting:

RSVD is a 5–6× win in 2-site DMRG, only.. For dmrg2-gpu at $\chi = 128$, randomized SVD reduces wall time from 124.6 s (baseline) to 22.7 s (5.49×); at $\chi = 256$, from 219.1 s to 34.9 s (6.29×). This is exactly the regime where Halko–Martinsson–Tropp pays off: in 2-site DMRG the per-bond matrix has full_k $= d \cdot \min(\chi_L, \chi_R)$, which exceeds the truncation rank $\chi_{\max}$ by a factor of d. The RSVD pipeline (project to $\chi_{\max} + p$ rank, QR, project back, small SVD, expand) consequently does asymptotically less work than the deterministic SVD it replaces. In contrast, single-site DMRG has full_k $= \min(\chi_L \cdot d, \chi_R) \leq \chi_{\max}$, so the RSVD guard never fires and the flag is a no-op (the dmrg-gpu row above).

HIP graph capture is a 1.2–1.8× win in pdmrg-gpu, only.. For pdmrg-gpu at $\chi = 128$, capturing `apply_heff` as a HIP graph and replaying across Lanczos iterations reduces wall time from 55.3 s to 47.0 s (1.18×); at $\chi = 256$, from 226.7 s to 128.4 s (1.77×, the largest LANCZOS_GRAPH win in the framework). The win is largest in pdmrg-gpu because its segment-parallel architecture launches many short-lived per-segment kernels whose individual launch latency is a substantial fraction of per-iteration wallclock; graph replay amortizes that latency over the captured subgraph. The same flag is flat in dmrg-gpu (where each Lanczos iteration already does enough work per launch) and outright incompatible with the Block-Davidson variants (`-gpu-opt`), where the captured graph would lock in the first-iteration subspace pointer and replay against stale columns; the variant constructors detect this and force the flag off, with the table cells marked † to distinguish from a bare regression.

Pattern.. The flags that target the *measured* bottleneck of a specific variant deliver wins; flags that try to redistribute work without changing what is bottlenecked do not. RSVD wins because the 2-site SVD has more singular values than the truncation needs. LANCZOS_GRAPH wins because pdmrg-gpu's per-segment launch overhead is a real fraction of wallclock. Neither win is portable to other variants: RSVD's mathematical regime does not exist in 1-site DMRG, and the launch-amortization regime does not apply where the per-call work is already large enough. This pattern—wins are variant-specific and bottleneck-specific—reinforces the broader Section 6 argument that GPU optimization for DMRG is a search for the locally-binding constraint, not a general algorithmic move.

5.6. The reveal: SVD profiling

Profiling of dmrg2-gpu at $L = 64$, $\chi = 256$ reveals where the wall time actually goes:

Table 13: Per-sweep wall-time breakdown for dmrg2-gpu, Heisenberg $L = 64$, $\chi = 256$.

Component	Time	Fraction
SVD (CPU LAPACK)	38–41 s	97–98%
Lanczos + H_{eff}	0.4–0.6 s	1–2%
Environment updates	0.03 s	$< 0.1\%$

The SVD operates on matrices of size $(\chi d, d\chi) = (512, 512)$ at $\chi = 256$ for the Heisenberg model. Each LAPACK `dgesvd` call takes ∼300 ms, and ∼126 SVDs per sweep (63 bonds × 2 directions) total ∼38 s.

This single number explains every negative result in the preceding sections. Newton–Schulz and Block-Davidson targeted the Lanczos eigensolver and the SVD factorization, but the eigensolver accounts for only 1–2% of wall time; even making it infinitely fast would be imperceptible. Newton–Schulz does target SVD, but replaces one $O(n^3)$ factorization with up to 30 GEMM iterations at $O(n^3)$ each—a 30× FLOP increase that the GPU's throughput advantage cannot offset at $n \leq 512$. Cross-segment batching targeted GEMM dispatch overhead in the eigensolver, again the wrong 2% of the runtime. Parallel DMRG's serial coupling sweeps each pay the full SVD cost per bond. And A2DMRG's TT-SVD compression—itself an $O(L\chi^3)$ SVD chain—is the very operation that dominates wall time.

GPU SVD via rocsolver provides a modest 13% improvement at $\chi = 256$ ($L = 64$: 124 s vs. 142 s for 3 sweeps), but paradoxically makes parallel DMRG *slower* because the smaller effective bond dimensions within segments ($\chi_{\text{eff}} \approx 50$–60) put the GPU SVD in a regime where its overhead exceeds the CPU's.

The profiling picture is unambiguous: at moderate bond dimensions, DMRG on a GPU is not a compute-bound problem. It is an SVD-bound problem, and the SVD runs on the CPU.

6. Discussion

The results above document a series of independent optimization attempts—GPU GEMM acceleration, two parallelization strategies, three alternative

algorithms—that all fail to deliver consistent speedups. Rather than analyzing each failure in isolation, we now unify them under three structural explanations.

6.1. The small-matrix wall

At $\chi \leq 50$, three distinct overhead mechanisms conspire to make the GPU slower than a single CPU core.

First, *GPU launch overhead*: each rocBLAS GEMM call incurs ∼5–10 μs of kernel dispatch latency, and a single Lanczos iteration requires hundreds of such calls. At $\chi = 50$, the arithmetic per GEMM (a 50×50 multiply taking ∼0.25 μs at peak) is smaller than the launch overhead itself, so the GPU spends more time dispatching work than performing it.

Second, *CPU threading overhead*: the same small matrices that starve the GPU also penalize multi-threaded CPU BLAS. Thread synchronization costs exceed the arithmetic savings, which is why 93% of CPU wins use a single thread. At these sizes, a single core streaming through L2-resident data is the most efficient executor available.

Third, *iterative algorithm overhead*: Newton–Schulz, Block-Davidson, and Chebyshev filtering all increase the total number of matrix operations per bond. When each operation is already overhead-dominated, multiplying the operation count makes the situation strictly worse.

These three effects are facets of the same underlying cause: at $\chi \leq 50$, DMRG's per-bond linear algebra problems are too small to amortize any form of parallelism overhead, whether hardware (GPU wavefronts, CPU threads) or algorithmic (iterative refinement loops).

6.2. The SVD ceiling

At $\chi \geq 128$, the GPU's GEMM throughput does overcome launch overhead for the Lanczos eigensolver—but this victory is pyrrhic. The profiling in Section 5.6 shows that SVD consumes 97–98% of per-sweep wall time at $\chi = 256$, imposing a ceiling that no eigensolver or environment-update optimization can breach.

The ceiling is structural. DMRG requires a full SVD (all singular values and vectors) at every bond to truncate the MPS, and LAPACK's QR-iteration algorithm [23] performs this in a single $O(n^3)$ pass with low constant factors. Any iterative replacement must either match this efficiency—difficult, since QR-iteration exploits dense matrix structure that iterative

methods do not—or accept a higher total FLOP count and rely on hardware throughput to compensate. Newton–Schulz attempts the latter strategy but requires up to 30 GEMM iterations, a $\sim$30$\times$ FLOP increase, and at $n = \chi d \leq 512$ the matrices remain below the $\sim$2048 threshold where the MI300X's MFMA units reach peak occupancy.

GPU SVD via rocsolver follows the same QR-iteration algorithm as LAPACK but adds device overhead (kernel launches, memory management) that exceeds the compute savings at $\chi \leq 256$. The 13% improvement observed at $\chi = 256$ suggests that the crossover for GPU SVD lies near $\chi \approx 300$–500.

6.3. The A2DMRG paradox

A2DMRG [2] illustrates a subtler failure mode: the algorithm is mathematically sound but operationally self-defeating. The warm-up experiment (Table 8) shows that each serial DMRG2 sweep buys 2–3 orders of magnitude in A2DMRG accuracy. By 5 warm-up sweeps the parallel phase converges in 1–2 iterations—but by then serial DMRG2 has already nearly converged, rendering the parallel phase redundant.

The underlying issue is that A2DMRG's additive (Jacobi-like) correction cannot compete with standard DMRG's multiplicative (Gauss–Seidel-like) sweep for sequential convergence rate. Parallelism could compensate if the per-iteration cost were sufficiently reduced, but A2DMRG's linear combination phase inflates the bond dimension to $(L-1) \times \chi$, and the subsequent TT-SVD compression—an $O(L\chi^3)$ serial SVD chain—is both slow and lossy. At $L = 32$, $\chi = 20$, the 31:1 compression ratio destroys the information that the parallel local solves produced.

This is not merely an implementation issue. The algorithm's design requires that the compression ratio grow linearly with L, creating a fundamental tension between system size and accuracy. The original paper [2] operates in a regime ($d \leq 24$, rank-growth initialization) where this tension is manageable; our lattice-model tests expose the scaling limit.

6.4. When would GPU DMRG win?

The three structural barriers—small-matrix overhead, SVD ceiling, and compression loss—point to the conditions under which GPU DMRG would provide clear advantages:

Large bond dimensions ($\chi \gtrsim 1024$).. At large χ, both the GEMM and SVD matrices exceed the size thresholds where GPU hardware reaches high occupancy. Newton–Schulz's 30× FLOP overhead would need to be offset by a > 30× GPU/CPU throughput ratio, which the MI300X can achieve only at $n \gtrsim 2048$. The numerical stability of Newton–Schulz at large condition numbers remains a concern.

Multi-GPU parallelism.. The real-space parallel DMRG algorithm [1] was designed for distributed-memory systems with one segment per processor. On a single GPU, the segment parallelism competes for the same compute resources. With multiple GPUs, each segment would have dedicated compute capacity, and the Amdahl's law bottleneck from serial coupling sweeps could be addressed through pipelining.

Alternative SVD approaches.. Randomized SVD [24] provides large speedups (5–6× at $\chi = 128$–256) in two-site DMRG, where the per-bond matrix has d times more singular values than the truncation retains; the ablation results in Section 5.5 confirm this. The same flag is a no-op in single-site DMRG, where the matrix never has more singular values than $\chi_{\max}$. Mixed-precision approaches (single-precision for warmup, double for final convergence) could halve the SVD time for the majority of sweeps and remain unexplored in this study.

Launch-overhead amortization for parallel variants.. HIP graph capture of `apply_heff` (Section 5.5) yields a 1.8× speedup at $\chi = 256$ in pdmrg-gpu, where the segment-parallel architecture launches many short-lived per-segment kernels. For variants whose per-call work is already large enough to amortize launch latency—dmrg-gpu, dmrg2-gpu—the same flag is flat. This is the second instance of the broader pattern: micro-optimizations that target the measured local bottleneck of a specific variant pay off; those that attempt to redistribute work without addressing the bottleneck do not.

The honest summary is that GPU DMRG at $\chi \leq 256$ is a solution in search of a problem large enough to justify it—except in two specific niches: two-site DMRG (where RSVD is binding) and segment-parallel pdmrg-gpu (where launch latency is binding).

7. Conclusions

We have presented a systematic benchmark of eleven DMRG implementations on the AMD MI300X GPU, spanning three physics models and a

range of system sizes and bond dimensions. Our main findings are:

1. The GPU vs. CPU crossover depends on both χ and L. At $\chi = 128$, the GPU wins 100% of short-chain ($L \leq 20$) configurations but only 71% at $L = 32$–64 and 50% at $L \geq 100$. At $\chi \leq 50$, the CPU wins all medium-to-large systems ($L \geq 32$).
2. When the CPU wins, it almost always wins with a single thread: 93% of CPU victories use single-threaded OpenBLAS. At $\chi \leq 50$, the matrices are small enough that thread synchronization overhead exceeds any parallel speedup, making a single core streaming through L2 cache the most efficient executor.
3. Two-site DMRG is $\sim$3$\times$ faster than single-site at large systems ($L = 100$) due to faster convergence from adaptive bond dimension growth.
4. Replacing BLAS-2 algorithms (Lanczos, LAPACK SVD) with BLAS-3 alternatives (Newton–Schulz polar decomposition, Block-Davidson, Chebyshev filtering) uniformly fails to improve performance at $\chi \leq 256$. The Newton–Schulz variant achieves a 0% win rate across 50 configurations and diverges numerically at $\chi \geq 128$. Chebyshev filtering is 1.9–11$\times$ slower. Cross-segment batched GEMM is slower in 18 of 19 configurations.
5. A systematic ablation of six GPU micro-optimizations across all six `-gpu` and `-gpu-opt` variants finds two real wins and four flat outcomes (Section 5.5). Randomized SVD yields 5–6$\times$ speedups in two-site DMRG, where the per-bond matrix has d times more singular values than the truncation retains. HIP graph capture of `apply_heff` yields 1.2–1.8$\times$ in segment-parallel pdmrg-gpu, where launch latency is a meaningful fraction of per-iteration wallclock. Both wins target the locally-binding bottleneck of a specific variant; the same flags are flat where their target bottleneck does not bind. Device-side truncation, fused Lanczos kernels, sparse-MPO compaction, and MFMA-friendly bond-dim padding are all flat across variants.
6. CPU SVD consumes 97–98% of per-sweep wall time at $\chi = 256$. This bottleneck cannot be addressed by any optimization that targets the remaining 2–3% of runtime.
7. Parallel DMRG on a single GPU is slower than serial two-site DMRG at all tested sizes, limited by the serial coupling sweep required to maintain global MPS consistency.

8. The additive two-level parallel DMRG (A2DMRG) algorithm [2] achieves machine-precision accuracy for $L \leq 20$ but degrades to 10^{-5} errors at $L = 64$ due to TT-SVD compression loss. With $P = 2$ ranks, it is 2–4× slower than serial DMRG2, and sufficient serial warm-up renders the parallel phase redundant. This is the first independent reproduction of the algorithm and the first test on lattice spin models.

Practitioner bottom line.. For production DMRG at $\chi \leq 256$, use quimb (or equivalent) with single-threaded BLAS on CPU. Move to GPU single-site DMRG only at $\chi \geq 128$ on short chains ($L \leq 20$), and to GPU two-site DMRG when convergence is slow at large L. Do not invest in parallelism or iterative SVD replacements at these bond dimensions.

These negative results delineate the boundary between problem regimes where GPU algorithmic variants can and cannot compete with mature CPU linear algebra. At moderate bond dimensions ($\chi \leq 256$), LAPACK's optimized SVD and the Lanczos algorithm's efficiency for single-eigenvalue problems make them difficult targets. The regime where BLAS-3 replacements would be expected to help—$\chi \gtrsim 1024$ on multi-GPU systems—remains to be explored. More broadly, our experience suggests that porting an algorithm to a GPU is not merely a matter of mapping operations to hardware: the structure of the algorithm itself—how much serial factorization it requires, how small its inner matrices are, how lossy its compression steps become—determines whether the port can ever pay off.

All source code (11 implementations in Python, Python/MPI, and C++/HIP), benchmark scripts, and raw benchmark data are available at `https://github.com/yye00/dmrg-implementations`.

CRediT authorship contribution statement

Yaakoub El Khamra: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The source code, benchmark scripts, and raw benchmark data are available at `https://github.com/yye00/dmrg-implementations`. The repository includes build instructions for ROCm 7.2 on AMD MI300X (gfx942). Benchmark results are stored in JSON format in the `benchmarks/paper_results/` directory, comprising 604 entries in the main dataset (`results.json`) and 108 entries in the GPU optimization study (`gpu_opt_bench.json`).

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Appendix A. Complete benchmark results

Tables A.14–A.16 report wall times (seconds) for all converged benchmark configurations. For CPU implementations (Q-D1: quimb-dmrg1, Q-D2: quimb-dmrg2), the reported time is the best across thread counts (1, 2, 4, 8, 12). For GPU implementations (D1-GPU: dmrg-gpu, D2-GPU: dmrg2-gpu, PD-GPU: pdmrg-gpu with $P = 2$ segments), all runs use a single MI300X with CPU SVD. The fastest implementation for each configuration is shown

in **bold**. A dash (—) indicates the implementation was not tested or did not converge for that configuration.

Table A.14: Complete wall times (seconds) for the Heisenberg spin-$\frac{1}{2}$ chain (OBC, $d = 2$, $D = 5$, real). Bold indicates fastest.

L	χ	Q-D1	Q-D2	D1-GPU	D2-GPU	PD-GPU
12	20	0.81	1.10	0.67	**0.52**	0.71
12	50	0.78	1.06	**0.42**	0.56	0.65
12	128	0.83	0.99	**0.42**	0.54	0.66
20	20	1.11	1.12	2.31	**1.07**	1.89
20	50	**1.08**	1.43	1.42	1.45	2.26
20	128	2.05	4.82	**1.36**	2.36	3.36
32	20	**1.50**	1.69	5.06	2.27	—
32	50	**1.82**	2.91	6.20	2.91	—
32	128	5.24	15.24	**4.24**	5.67	—
64	50	5.67	**5.05**	26.99	9.27	24.50
64	128	**20.59**	37.69	24.01	22.80	—
100	50	11.36	**8.50**	74.22	43.09	—
100	128	**40.70**	75.68	149.40	53.57	—

The full results reinforce the narrative developed in Sections 5–6:

- **GPU single-site (D1-GPU) dominates the TFIM**: it wins 10 of 13 TFIM configurations, including all of $L \leq 20$. The TFIM's small MPO bond dimension ($D = 3$) and fast convergence (2–3 sweeps) keep per-bond costs low, favoring the GPU even at moderate χ—precisely the small-matrix regime where low per-bond overhead matters most (Section 6.1).

- **Josephson is hardest for the GPU**: complex arithmetic doubles the GEMM cost, and $d = 5$ inflates the two-site eigensolve dimension to $\chi d^2 = 25\chi$. D2-GPU is never the fastest for Josephson at $\chi \geq 50$. The CPU (Q-D1) wins all Josephson configurations at $\chi = 50$, and the GPU only overtakes at $\chi = 128$ for $L \leq 32$—consistent with the SVD ceiling becoming the binding constraint once the eigensolver is fast enough (Section 6.2).

Table A.15: Complete wall times (seconds) for the Josephson junction array (OBC, $d = 5$, $D = 4$, complex). Bold indicates fastest.

L	χ	Q-D1	Q-D2	D1-GPU	D2-GPU	PD-GPU
8	20	1.46	**1.31**	2.67	1.49	4.70
8	50	1.25	3.18	**1.10**	1.16	19.05
8	128	3.55	6.52	**0.97**	2.54	20.01
16	20	**2.30**	3.53	6.19	6.38	12.40
16	50	**3.73**	16.32	10.70	8.03	55.61
16	128	26.19	228.10	**4.67**	29.17	314.34
32	50	**17.07**	59.69	32.30	126.23	239.93
32	128	107.98	891.90	**33.02**	107.87	800.42
48	50	**31.00**	360.51	64.69	244.94	326.85
48	128	**172.77**	—	191.16	1347.92	—
64	50	**41.62**	1287.38	88.55	568.37	—
64	128	300.79	—	**295.82**	981.48	—

- **Parallel DMRG (PD-GPU) never wins**: across all 28 configurations where it was tested, pdmrg-gpu is always slower than both the best serial GPU and best CPU implementation. Its closest approach to competitiveness is Heisenberg $L = 12$, $\chi = 20$ (0.71 s vs. 0.52 s for D2-GPU), but even there it loses. Every parallel segment sweep still pays the full per-bond SVD cost, and the serial coupling phases add to that baseline.

- **Large Heisenberg systems favor CPU or two-site GPU**: at $L \geq 64$, the CPU (Q-D1 or Q-D2) wins at $\chi \leq 128$. When the GPU wins (e.g., $L = 32$, $\chi = 128$), it is always D1-GPU. D2-GPU is competitive at $L = 64$, $\chi = 128$ (22.80 s vs. 24.01 s for D1-GPU) but does not win—the two-site SVD cost ($d^2\chi$ matrices) pushes it further into the SVD-dominated regime.

Table A.16: Complete wall times (seconds) for the transverse-field Ising model at criticality (OBC, $d = 2$, $D = 3$, real). Bold indicates fastest.

L	χ	Q-D1	Q-D2	D1-GPU	D2-GPU	PD-GPU
12	20	0.77	0.79	**0.39**	0.51	5.18
12	50	0.78	0.80	**0.41**	0.49	4.54
12	128	0.83	0.84	**0.41**	0.52	1.38
20	20	0.88	1.00	**0.69**	1.01	12.41
20	50	1.03	1.20	**0.98**	1.76	16.46
20	128	1.95	1.87	**1.52**	3.47	13.48
32	20	**1.14**	1.26	1.50	1.50	18.71
32	50	**1.64**	1.83	1.78	3.78	29.75
32	128	5.81	6.17	**4.89**	13.46	45.78
64	50	**3.97**	4.80	4.96	8.67	27.36
64	128	22.11	26.59	**20.45**	51.75	45.55
100	50	**7.66**	9.31	8.36	15.97	59.89
100	128	40.81	55.31	**36.31**	96.98	—